Faculty of Mathematics and Informatics
Computer Science Department

Multimedia Applications & Virtual Reality
Master : Artificial Intelligence & Applications (AIA)
Session 1 Exam –

The standard answer key:
Correction elements

Exercise 1 : Check the correct answer(s): A wrong choice cancels the answer **(06 pts)**

0.25 / Correct Choice

1	2. A computer generated virtual world:	*May or may not be a simulation of concepts in physical world*	
		Has no relation to simulation	
		Has no relation to the physical world	
		Is always a simulation of concepts in physical world	X
2	Immersion can be defined as (Check the complete definition)	*A transfer of perception without transferring consciousness*	
		A transfer of one or more perceptions	
		A transfer of all perceptions	
		A transfer of perception for the purpose of transferring consciousness.	X
3	The nature of contradiction interferences is	*A perception of the physical world*	
		A perception of the virtual world	
		A partial unintentional perception of the physical world	X
		A partial intentional perception of the physical world	
4	The visual display of the virtual world is carried out by	*Direct rendering of 3D instances*	
		Drawing precalculated 2D versions	
		Display of the face closest to the 3D object	
		The 3D instance projections on 2D screens depending on viewing angle	X
5	A site map (incidence and neighborhood between places and monuments) in a VW can serve as	*Kinematic model*	
		Analytical model associated with VW	
		Semantic model	
		Topological model	X
6	Modeling mountains and plateaus (terrains) involves	*Non-deformable rigids*	X
		Deformable rigids	
		Non-solids	
		Solids	
7	The resolution of geometric objects is	*Infinite by definition*	
		Finite by definition	X
		Finite except for vertex-based structures	
		Finished except for edge-based structures	
8	In a VR, immersion is implemented by :	*Creating physical stimuli for virtual entities*	
		Replacing physical stimuli with virtual stimuli	X
		Creating physical objects that represent virtual objects	
		Replacing physical objects with virtual objects	
9	Access to space in nonparametric analytical structures is	*Direct*	X
		Parametrized	
		Constant	
		Linear	
10	When visual perception is not exclusive to the virtual world	*It's a total immersion*	
		This is partial immersion or augmented reality	X
		This is contradiction interference reduced to zero	
		This is a total lack of immersion	

11 In practice, a device is said to be haptic when it :	*Implements a haptic perception*	
	Communicates with a virtual identity in the virtual world	X
	Implements communication with the virtual world in a haptic dimension	
	Communicates with the application through movements	
12 The separation of the stereoscopic image at the screen level implies that	*One of the images hides the other*	
	One of the images is encrypted and the other is not	
	Both images are displayed simultaneously	X
	None of the previous three answers	
13 Active glasses separation is a separation	*At screen level*	
	Requiring single-color filters	
	Requiring display on spectacle lenses	
	Which uses a remotely controlled mechanical device	X
14 When real objects are perceived while contradiction interference is zero	*User does not see visual display*	
	The user is in partial immersion	
	The user is in an augmented virtuality	X
	None of the previous three answers	
15 In theory, a device that communicates with the application through movements is called	*A haptic device*	X
	An immersive device	
	A VR device	
	A communication device	
16 Pneumatic return interfaces (micro-balloons) exploit sensitivity	*To the temperature*	
	To the pressure	X
	To the physical contact	X
	To the vibrations	
17 The force feedback in externally implemented interfaces is essentially	*Mechanical*	X
	Based on temperature sensitivity	
	Based on pressure sensitivity	
	Based on vibration sensitivity	
18 The choice of sound device types depends on	*The degree of freedom and the level of immersion desired*	X
	The diffused sound quality	X
	The amplitude of sounds in virtual scenes	
	None of the previous three answers	
19 Multichannel sound in a scene is defined:	*In the virtual scene, but in reality it is always two channels*	
	In the virtual scene with virtual sources	
	In the physical scene with one sound source for each channel	X
	In the virtual scene with one sound source for each channel	
20 Implementing a VR solution requires (check the complete answer):	*A game engine or 3D IDE, a scenario and behavioral interfaces*	X
	A game engine or 3D IDE, and if possible behavioral interfaces	
	A game engine or 3D IDE, a scenario and if possible behavioral interfaces	
	A game engine or 3D IDE	

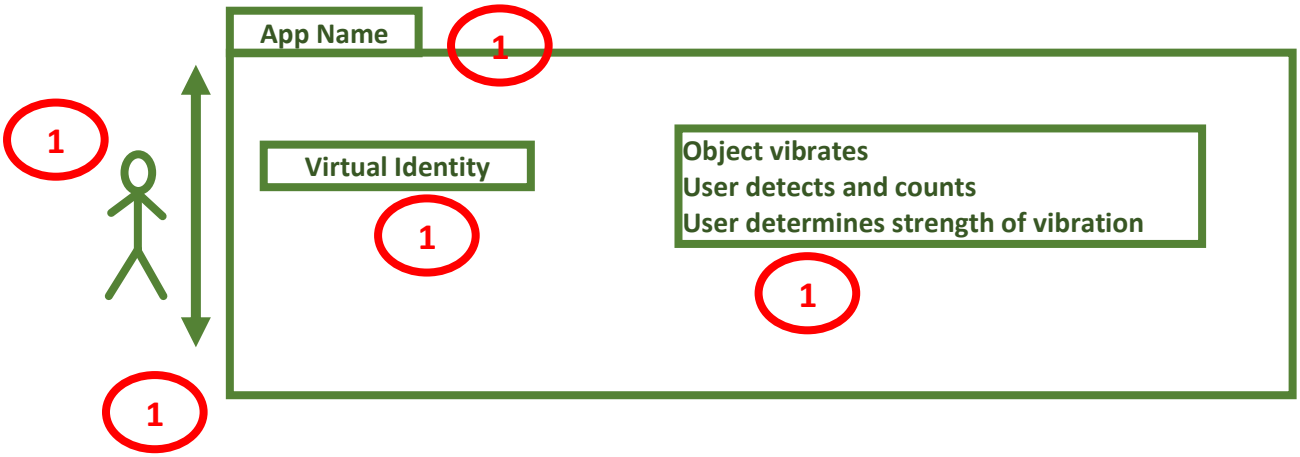
Exercise 2 : (07 pts)

Consider a simple full-immersive VR application in which the user is asked to count the number of vibrations of an object and specify the vibration strength (Strong or Weak)

1. What behavioral devices are suitable for such an application

HMD + Data Gloves 1 + 1

Propose an SDL diagram modeling such an application



Exercise 3 : (07 pts)

The following VRML code render a set of objects to recognize:

```
#VRML V2.0 utf8
WorldInfo{
         Title "Recognize objects"  }
 DEF FBOX Shape {
         appearance Appearance {
                 material Material{ transparency 0.5}  }
         geometry Box {
                size 4 4 4 }  }
 Transform [
     translation 1 1 -1
     children Shape {
          appearance Appearance {
                 material Material{} }
          geometry  Cylinder {
               bottom  TRUE
               height  2
               radius  1
               side    TRUE
               top     TRUE } } }
 Transform {
     translation 2 5 7
     Children Shape {
          appearance Appearance {
                 material Material{} }
          geometry  Cone {
               bottomRadius 2
               height  1
               side    TRUE
               bottom     TRUE } } }
 Transform {
     translation 2 5 7
     children Shape {
          appearance Appearance {
                 material Material{} }
          geometry  Sphere {
               radius 4 } }
 Transform {
     translation 9 8 -5
```

```
39.       children Shape {
40.            appearance Appearance {
41.                  material Material{} }
42.            geometry  Cylinder {
43.                 bottom  TRUE
44.                 height  3
45.                 radius  2
46.                 side    TRUE
47.                 top     TRUE } } }
```

1. Is there any bug or problem in the final rendering ?

   Line number : 03 Bug Description : Title 0.25 + 0.25

   Line number : 08 Bug Description : } missed 0.25 + 0.25

   Line number : 09 Bug Description : { missed 0.25 + 0.25

   Line number : 22 Bug Description :Children 0.25 + 0.25

   Line number : 36 Bug Description : } missed 0.25 + 0.25

Correct the previous code to render correctly on the screen

Line number : 03 | Correct code : Title "Recognize objects" } 0.25 + 02.5

Line number : 08 | Correct code : size 4 4 4 } } 0.25 + 0.25

Line number : 09 | Correct code : Transform [0.25 + 0.25

Line number : 22 | Correct code : Children Shape { 0.25 + 0.25

Line number : 36 | Correct code : radius 4 } } 0.25 + 0.25

2. Ho many objects will be displayed to the user to be identified ? **4** 01

3. How can you modify the parameters (Only scalar fields without adding new nodes) to increase the number of displayed objects ?

Line number : 21| Modified code : translation 8 10 20 01